

RAFAŁ ZAJĄC

FULL STACK DEVELOPER

Phone (+49) 1705214741

E-mail rf.zajac@tutamail.com

LinkedIn <https://www.linkedin.com/in/rafalzajac88/>

GitHub <https://github.com/RZajacc>

Place of residence **Berlin, Germany**



I'm a Full Stack Software Developer with commercial experience in React, TypeScript, NestJS, Node.js and MySQL. Currently work on modernising a legacy Java/JSP transaction platform, developing REST APIs, database structures and migrations, authentication and interactive analytics features.

My path into software development has been gradual and consistent. Before moving fully into engineering, I spent several years in academic research and six years in professional sports analytics. I hold a PhD in Physical Culture Sciences and worked with data in Germany's Bundesliga, the CEV Champions League and Poland's top volleyball division. This background shaped the way I work today: I'm highly analytical, data-driven and comfortable translating complex requirements into practical solutions. I have also used Python and VBA to automate repetitive workflows and gained hands-on experience with Docker, AWS, CI/CD and AI-assisted development. I'm based in Berlin, with professional English and advanced German currently progressing towards C1. I'm looking for Full Stack Software Developer opportunities in Germany, ideally in Berlin or in a remote environment.

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, Zustand, Material UI, Tailwind CSS, React Grid Layout, React Hook Form, Zod, Vite, HTML5, CSS3

Backend & APIs: Node.js, NestJS, REST API Development, Swagger / OpenAPI, JWT Authentication, Passport, Password & Token Management

Databases: MySQL, PostgreSQL, MongoDB, SQL, TypeORM, Mongoose, Database Schema Design, Relational Data Modelling, Database Migrations

Testing: Jest, React Testing Library, Unit Testing, Integration Testing

DevOps & Cloud: Docker, Docker Compose, Git, GitHub, GitHub Actions, CI/CD, AWS Lambda, API Gateway, S3, CloudFront, Route 53, Serverless Architecture

Additional Technologies: Java, Spring Boot, Python, VBA, Next.js, Figma, DBeaver

AI-Assisted Development: Claude, JetBrains AI Assistant, AI-assisted debugging, API documentation, solution brainstorming and code adaptation

PROFESSIONAL EXPERIENCE

Full Stack Software Developer / Sensorfarm Polska

April 2025 – Present • Berlin, Germany

Small software consultancy and product company developing custom applications and proprietary software products.

Transaction & Analytics Platform Modernisation (January 2026 – Present)

- Implementing the end-to-end migration of a legacy Java/JSP transaction platform into a modern React and NestJS application, working directly with the CEO/developer who defines the product requirements and key technical direction.
- Working across frontend, backend and database layers on an application currently exposing 53 REST API endpoints.

- Implemented an EAV-based data model designed to support multiple clients with different transaction structures, including the design of 4 relational database tables, relationships and migrations.
- Designed and built a configurable analytics dashboard with React Grid Layout, enabling users to arrange, resize and manage interactive data visualisations.
- Implemented secure JWT-based authentication, refresh token rotation, password hashing and email-based self-service password recovery.
- Introduced frontend unit and integration testing with Jest and React Testing Library, reaching 57% line coverage in the tested areas.
- Expanded the product from a client-specific transaction application towards a configurable multi-client platform with custom data structures, self-service account management and interactive analytics.

Stack: React, TypeScript, NestJS, Node.js, MySQL, TypeORM, Docker, Jest, React Testing Library

Operations Management Platform (April 2025 - December 2025)

- Contributed for nine months to an established map-based operations management platform built on a microservices architecture, working primarily on frontend development.
- Designed an operations dashboard in Figma for grouping and presenting map-based activities and subsequently implemented the approved solution in React and Material UI.
- Developed frontend functionality for managing and visualising operational data using React and Zustand and integrating with backend services.
- Contributed selectively to Java/Spring Boot backend and PostgreSQL database tasks outside the primary frontend scope.

Stack: React, TypeScript, Zustand, Material UI, Java, Spring Boot, PostgreSQL, Docker

Professional Development & Independent Software Development

January 2024 – March 2025 • Berlin, Germany

Following completion of an intensive full-stack development programme at Code Academy Berlin, focused on building and extending independent web applications while deepening practical skills in **React, TypeScript, Node.js, Next.js and software testing**. Complemented project-based development with specialised technical courses and completed German language training at B2 level.

Full Stack Web Application - CodaSk *React, TypeScript, Tailwind CSS, NestJS, MongoDB, Mongoose, Docker*

- Took further ownership of a full-stack educational platform originally developed as a collaborative bootcamp project, continuing its development independently after the original team stopped actively contributing.
- Expanded and refined frontend and backend functionality around user accounts, questions and answers, search, filtering and profile management.
- Used the project to deepen practical experience in full-stack application architecture, authentication, API development and data handling.
- Later deployed the application on AWS using S3, CloudFront, Lambda, API Gateway and Route 53, with automated frontend and backend deployments through GitHub Actions.

Recipe Discovery Application- ForkIt *React, TypeScript, React Router, Firebase, Vite, Sass, Spoonacular API*

- Developed and extended a web application for discovering and saving recipes using the Spoonacular API.
- Implemented user-facing functionality including recipe discovery, categorisation, user accounts and favourites.
- Strengthened practical experience with React, TypeScript, external API integration and Firebase.

Developer Portfolio, rfzajac.dev *Next.js, TypeScript, Tailwind CSS, Vercel*

- Designed, built and deployed a personal developer portfolio to present technical projects, skills and professional experience.
- Used the project to gain additional hands-on experience with Next.js, Tailwind CSS and Vercel deployment.

Performance Data Analyst & Volleyball Statistician / Berlin **Recycling Volleys**

August 2019 – May 2023 • Berlin, Germany

Professional volleyball club competing in the German Bundesliga and CEV Champions League.

- Collected, processed and analysed match and opponent data using Data Volley to support game preparation and tactical decision-making.
- Worked closely with the coaching staff to identify analytical needs, define relevant performance indicators and translate data into actionable match strategies.
- Prepared and presented opponent analysis and game recommendations within short 3-6 day competition cycles.
- Developed Python scripts to automate repetitive data-cleaning tasks across approximately 10 analysis files per week, each containing 800-1,500 rows.
- Reduced a recurring data-preparation process from around 60 minutes of manual work to less than 10 minutes, improving the efficiency of the weekly analytical workflow.
- Continuously refined analytical and presentation methods based on coaching feedback and changing competition requirements.

Performance Analyst & Volleyball Statistician / MKS Dąbrowa **Górnica**

August 2017 – September 2019 • Dąbrowa Górnicza, Poland

Professional women's volleyball club competing in Poland's top division and subsequently in the 1st League.

- Collected and analysed match and opponent data using Data Volley to support tactical preparation and match strategy.
- Prepared game-specific analyses and presented findings to the coaching staff and team.
- Adapted analytical methods and reporting to the needs of coaches and the requirements of individual opponents.
- Used Excel and VBA to streamline recurring reporting tasks, reducing selected preparation activities from around 15 minutes to under 5 minutes.

Researcher & Senior Technician / The Jerzy Kukuczka Academy of **Physical Education in Katowice**

September 2016 – June 2017 • Katowice, Poland

Academic research career developed alongside doctoral studies in Physical Culture Sciences (Sport Science), progressing from research and teaching responsibilities to a role focused primarily on experimental research, data analysis and independent project ownership.

- Managed selected research projects end to end, from defining research questions and identifying study groups through data collection, preprocessing, analysis and presentation of findings.
- Worked with experimental and biomechanical data using Excel and MATLAB-based analytical tools.

- Conducted research into human balance and postural control, including studies involving competitive athletes and comparison groups.
- Had autonomy in selecting research topics, analytical methods and approaches for selected projects.
- Contributed to doctoral research later published in **PLOS ONE**, with responsibilities including data curation, formal analysis, investigation, methodology, software and validation.
- Presented research findings at the international **Motor Control Conference in 2012**.
- Supported onboarding of junior colleagues in laboratory procedures, research equipment and supporting software.
- Combined research and teaching responsibilities before moving into a technical role focused primarily on research and data analysis.

Selected Research & Academic Contributions

Biathletes present repeating patterns of postural control to maintain their balance while shooting

PLOS ONE, 2022 Co-author

Peer-reviewed research based on doctoral work examining postural control and balance in competitive biathletes. Contributed to data curation, formal analysis, investigation, methodology, software and validation.

Motor Control Conference 2012

Presented research findings during the poster session at an international scientific conference.

EDUCATION

Full Stack Web Development, Code Academy Berlin 2023

Intensive 840-hour, project-based programme covering full-stack web development with a focus on the MERN ecosystem, collaborative development and Scrum.

Computer Science Studies

University College of Enterprise and Administration in Lublin 2016-2018

Completed four semesters of part-time Computer Science studies alongside a professional career, with a strong focus on Java programming and software development fundamentals.

PhD in Physical Culture Sciences (Sport Science)

The Jerzy Kukuczka Academy of Physical Education in Katowice *Awarded 2017*

Doctoral Studies 2011-2016

MSc in Physical Education

The Jerzy Kukuczka Academy of Physical Education in Katowice *2009-2011*

BSc in Physical Education

The Jerzy Kukuczka Academy of Physical Education in Katowice *2006-2009*

LANGUAGES

- **English** - C1/Full Professional Proficiency
- **German** – Advanced, currently completing C1-level training
- **Polish** - Native

SOFT SKILLS

- Analytical Problem Solving
- Independent Ownership
- Requirements Analysis
- Data-Driven Decision Making
- Stakeholder Communication
- International Collaboration